# Threat Intelligence Platform: Labs 7, 8 and 9

Docker deployment and submission information · generated 2026-10-14 19:52 UTC

## Project description

A FastAPI / MongoDB threat-intelligence dashboard (Lab 7) running on a three-member MongoDB 7.0 replica set `rs0` (Lab 8), together with the Lab 9 write-concern latency benchmark and network-partition experiment. It is delivered as one application image plus the official `mongo:7.0` image, orchestrated by Docker Compose with persistent named volumes.

## Docker Hub

| | |
|---|---|
| **Docker Hub repository** | https://hub.docker.com/r/jovit206/threat-intelligence-platform |
| **Published tags** | latest |
| **Image digest** | `sha256:b6656e1cd864778243c72e66fe0f230876a83f2ab27da73f0571c99e27263031` |
| **Verified on Docker Hub** | 2026-10-14 19:52 UTC (manifest inspected; not pulled) |
| **Pull command** | `docker pull jovit206/threat-intelligence-platform:latest` |

## Start the complete application

```
cp .env.example .env
mkdir -p labs/lab9/results
# set APP_IMAGE=jovit206/threat-intelligence-platform:latest in .env, then:
docker compose pull
docker compose up -d
# open http://localhost:8000
```

Compose starts the three MongoDB members, initialises `rs0` only if needed (idempotent), then starts the application. Data lives in named volumes and survives `stop`, `down` and upgrades.

## Features

- **Lab 7**: Live Threat Stream, Top Threats, Report Corpus, Search & Enrich (VirusTotal, 24 h cache), Cross-Source Analysis, Working Set Monitor (live WiredTiger cache telemetry).
- **Lab 8**: three-member replica set, member and primary monitoring, failover observer ( `labs/lab8` ).
- **Lab 9.1**: write-concern latency for `w:1` , `w:majority` , `w:all` (numeric `w` = member count) with `j:false` / `j:true` , 1,000 inserts each; JSON/CSV artifacts; historical vs live results separated in the UI.
- **Lab 9.2**: confirmed, reversible network-partition simulation with prechecks, timestamps, honest write outcomes and verified recovery.

## Lab 9.1 experiment (manual)

```
docker compose exec lab7-backend python labs/lab9/lab_01_write_concern_latency.py
# options: --docs 200   --dry-run   --replace-historical
```

Reports are written to `labs/lab9/results/lab_01_<timestamp>.json` and `.csv`. MongoDB has no `w: "all"`; it is run as `w:3` and the mapping is printed and saved.

## Lab 9.2 experiment (manual)

```
python labs/lab9/lab_02_simulate_partition.py --dry-run   # prechecks only
python labs/lab9/lab_02_simulate_partition.py             # type the confirmation text when
asked
```

Report: `labs/lab9/results/lab_02_partition_<timestamp>.json`. Losing the majority is a separate, separately confirmed phase (`--phase majority-loss`).

## Safety notes

- The partition experiment and the live benchmark never run automatically (build, start-up, tests, publication).
- Only a secondary is isolated by default; the primary is never selected. Containers, volumes, replica-set configuration and documents are not removed.
- `docker compose down -v` deletes database volumes and is not part of any procedure.
- No secrets are in the image: `.env` is excluded; the VirusTotal key is read by the backend at run time only.
- MongoDB is bound to loopback host ports and runs without authentication inside the internal network.

## Verification summary

Recorded 2026-10-15. Results are copied from the commands actually run; items marked MANUAL require a person to run them.

| Check | Result | Detail |
|---|---|---|
| Unit tests (81): backend, feeds, galaxies/reports, reliability, VirusTotal, write concern, partition script | **PASS** | python -m unittest <7 modules> |
| Lab 9.2 logic against a fake docker cluster (prechecks, confirmation, cleanup, reconnect fallbacks, report JSON) | **PASS** | 18 tests; no real docker used |
| Lab 9.2 read-only --dry-run against the real replica set | **PASS** | prechecks passed; containers/network unchanged |

| Check | Result | Detail |
| --- | --- | --- |
| Lab 9.2 real partition experiment | **MANUAL** | not executed in this task by instruction; run python labs/lab9/lab_02_simulate_partition.py |
| Lab 9.1 benchmark | **MANUAL** | ran earlier on a disposable stack (1000 x 6 acknowledged); re-run on your cluster when presenting |
| Frontend files byte-identical before/after migration | **PASS** | 6 files sha256 |
| OpenAPI contract identical (32 operations) | **PASS** | methods, parameters, status codes |
| Browser DOM + computed-style fingerprint, 8 views (real Edge) | **PASS** | identical before/after |
| Integration script against migrated image and live app | **PASS** | / , APIs, markup checks |
| docker compose config | **PASS** | valid |
| Existing MongoDB data | **PASS** | 21 collections, none removed, none with fewer documents |
| Image republished to Docker Hub | **MANUAL** | not done; published latest predates this work |